

Business Intelligence Capstone Report Outline

Project Title

AI-Assisted Business Intelligence Analysis of Indian Crop Production (1997–2015)

1. Executive Summary

Content Plan

- Brief overview of the project objective.
 - Introduction to the Indian Crop Production dataset.
 - Summary of the dashboard and Business Intelligence approach.
 - Highlight the three key insights discovered.
 - Present the main recommendations.
 - Conclude with the business value of using BI and AI in agricultural decision-making.
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2. Business Context

Content Plan

- Importance of agriculture in India's economy.
 - Challenges in understanding agricultural production across states, crops, and seasons.
 - Need for data-driven decision-making in agriculture.
 - Role of Business Intelligence in transforming raw agricultural data into actionable insights.
 - Purpose of the project and intended stakeholders, including policymakers and agricultural planners.
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3. Data Sources

Content Plan

- Dataset used: Indian Crop Production Dataset.
 - Source of the dataset.
 - Time period covered: 1997–2015.
 - Dataset size: 238,838 records.
 - Key variables:
 - State
 - District
 - Crop
 - Season
 - Year
 - Area
 - Production
 - Yield
 - Data cleaning activities:
 - Removed duplicates.
 - Handled missing values.
 - Verified data types.
 - Dataset limitations:
 - No weather data.
 - No soil quality information.
 - No market price data.
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4. Key Metrics / KPIs

Content Plan

Describe each KPI and its business relevance.

- Total Crop Production
- Total Cultivated Area
- Average Yield
- Number of Crop Types
- Production by Season
- Production Trend Over Time
- Top Producing Crops

Explain how these KPIs help monitor agricultural performance and support decision-making.

5. Key Insights

Insight 1

Topic

Crop production is concentrated among a few major crops.

Supporting Chart

Top 10 Crops by Production

Discussion

- Explain production concentration.
 - Identify the highest-producing crops.
 - Business implication.
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Insight 2

Topic

Agricultural production changes over time.

Supporting Chart

Production Trend by Year

Discussion

- Long-term production trend.
 - Year-to-year fluctuations.
 - Importance for planning.
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Insight 3

Topic

Seasonal production patterns influence agricultural output.

Supporting Chart

Production by Season

Discussion

- Compare seasonal production.
- Importance of seasonal planning.

- Resource allocation implications.
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6. Strategic Recommendations

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Recommendation 1

- Continue supporting high-performing crops.
- Promote sustainable farming practices.

Recommendation 2

- Increase investment in lower-performing crops.
- Improve irrigation and farmer training.

Recommendation 3

- Expand Business Intelligence dashboards.
- Integrate AI for predictive agricultural analytics.
- Include weather, soil, and market-price data in future analyses.

Explain the expected benefits of each recommendation.

7. Limitations & Future Work

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Current limitations

- Historical dataset only.
- Limited variables.
- No real-time updates.
- No external environmental factors.

Future improvements

- Real-time agricultural monitoring.
- Weather integration.
- Soil quality analysis.

- Crop price forecasting.
 - Machine Learning-based yield prediction.
 - Satellite imagery integration.
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8. Appendix

Content Plan

Include the following supporting materials:

- Google Looker Studio Dashboard Link
 - Week 6 Insight Deck
 - Week 7 Prompt Log
 - AI Validation Log
 - Dataset Source
 - References used throughout the project
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Report Summary

Section	Target Word Count
Executive Summary	250
Business Context	200
Data Sources	150
Key Metrics / KPIs	200
Insight 1	250
Insight 2	250
Insight 3	250
Strategic Recommendations	300
Limitations & Future Work	150